Hands-On Lab: Prompt Engineering – Prompt Types Explained + Exercises

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1️⃣Instruction-Based Prompt – Definition + Hands-On

Definition:

An instruction-based prompt directly tells the AI what task to perform. It is the most basic form of prompting and forms the foundation for all other types.

Trainer Explanation:

Instruction prompts are straightforward. The AI behaves according to the clarity and structure of your command. The better the instruction, the better the result.

Hands-On Exercise:

1. Type the following prompt into ChatGPT: **“Explain Docker.”**
2. Now refine it: **“Explain Docker to a beginner using a real-world example in under 5 lines.”**

Expected Outcome:

The refined prompt produces a clearer, more structured, and beginner-friendly output.

Trainer Explanation:

Instruction prompts are the simplest form of prompting. You directly tell the model what task you want it to perform.

Hands-On Exercise:

1. Type the following prompt into ChatGPT: **“Explain Docker.”**
2. Now refine it and observe the difference: **“Explain Docker to a beginner using a real-world example in under 5 lines.”**

Expected Outcome:

The refined version should be simpler, clearer, shorter, and more beginner-friendly.

2️⃣Role-Based Prompt – Definition + Hands-On

Definition:

A role-based prompt assigns a persona or identity to the AI, such as a DevOps engineer, teacher, CEO, or doctor. This changes the tone, depth, and perspective of the response.

Trainer Explanation:

By giving the AI a role, you're guiding it to think and respond from a specific point of view. This instantly alters the style, vocabulary, and level of detail.

Hands-On Exercise:

Prompt 1: **"You are a senior DevOps engineer. Explain CI/CD pipelines to a junior developer."**

Prompt 2: **"You are a business consultant. Explain CI/CD from a business value perspective."**

Expected Outcome:

Two different perspectives for the same topic, aligned with the assigned roles.

Trainer Explanation:

Role prompts change the AI's tone, depth, and behavior by assigning it a specific identity.

Hands-On Exercise:

Prompt 1: **"You are a senior DevOps engineer. Explain CI/CD pipelines to a junior developer."**

Prompt 2: **"You are a business consultant. Explain CI/CD from a business value perspective."**

Expected Outcome:

Two completely different explanations for the same topic based on the assigned roles.

3️⃣Style-Based Prompt – Definition + Hands-On

Definition:

A style-based prompt controls the tone, emotion, or writing style of the AI's output. This can include formal, casual, storytelling, humorous, poetic, etc.

Trainer Explanation:

This prompt type is used when you want the AI to match a specific communication style. This is useful for presentations, storytelling, branding, and tailored communication.

Hands-On Exercise:

Prompt 1: **“Explain Kubernetes in a bedtime-story style.”**

Prompt 2: **“Explain Kubernetes in a formal tone suitable for a client executive meeting.”**

Expected Outcome:

One output will be imaginative and friendly; the other will be polished and professional.

Trainer Explanation:

You can change the response style (formal, fun, story-based, poetic, etc.).

Hands-On Exercise:

Prompt 1: **“Explain Kubernetes in a bedtime-story style.”**

Prompt 2: **“Explain Kubernetes in a formal tone suitable for a client executive meeting.”**

Expected Outcome:

One creative and playful output, one corporate and polished.

4️⃣Zero-Shot Prompt – Definition + Hands-On

Definition:

A zero-shot prompt provides no examples. You rely entirely on the model’s general knowledge and instruction-following ability.

Trainer Explanation:

This type is ideal when you want the AI to perform a task directly without needing pattern examples.

Hands-On Exercise:

“Write a user story for a shopping cart checkout feature.”

Expected Outcome:

A generic user story aligned to common Agile patterns.

Trainer Explanation:

The model performs the task without any examples.

Hands-On Exercise:

“Write a user story for a shopping cart checkout feature.”

Expected Outcome:

A general but structurally correct user story.

5️⃣Few-Shot Prompt – Definition + Hands-On

Definition:

Few-shot prompts give the model examples to learn from. The AI identifies the pattern and continues it.

Trainer Explanation:

Few-shot prompting is extremely powerful when you need formatted, structured, or stylistically consistent outputs.

Hands-On Exercise:

"Convert sentences into simpler English. Example 1: ‘The precipitation today is considerable.’ → ‘It is raining a lot today.’ Example 2: ‘The child was exhilarated.’ → ‘The child was very happy.’

Now simplify this: ‘The experiment yielded unexpected complexities.’"

Expected Outcome:

The model simplifies the sentence following the pattern in the examples.

Trainer Explanation:

You provide example outputs so the model learns the pattern.

Hands-On Exercise:

Prompt: "Convert sentences into simpler English. Example 1: 'The precipitation today is considerable.' → 'It is raining a lot today.' Example 2: 'The child was exhilarated.' → 'The child was very happy.'

Now simplify this: 'The experiment yielded unexpected complexities.'"

Expected Outcome:

The model should match the style and simplicity of the given examples.

6️⃣Chain-of-Thought Prompt – Definition + Hands-On

Definition:

A chain-of-thought prompt encourages the AI to show its reasoning process step by step.

Trainer Explanation:

This is ideal for math, logic, planning, and troubleshooting tasks where transparency matters.

Hands-On Exercise:

Prompt: **"How many 5-rupee coins are needed to make 145 rupees? Think step by step."**

Expected Outcome:

A detailed reasoning process accompanied by the correct calculation.

Trainer Explanation:

This prompt forces the AI to explain its reasoning.

Hands-On Exercise:

Prompt: **"How many 5-rupee coins are needed to make 145 rupees? Think step by step."**

Expected Outcome:

A detailed reasoning process followed by the correct answer.

7️⃣Constraint-Based Prompt – Definition + Hands-On

Definition:

Constraint-based prompts impose limits on the AI's response such as word count, tone, structure, or vocabulary restrictions.

Trainer Explanation:

Constraints force the AI to follow rules, producing consistent and controlled output.

Hands-On Exercise:

Prompt: **"Explain Terraform in exactly 4 bullet points using only simple English."**

Expected Outcome:

The output must follow your constraints: 4 points, simple English.

Trainer Explanation:

Constraints help control length, complexity, and structure.

Hands-On Exercise:

Prompt: **"Explain Terraform in exactly 4 bullet points using only simple English."**

Expected Outcome:

The response must follow your limits: 4 points, simple language.

8️⃣Delimiter-Based Prompt – Definition + Hands-On

Definition:

Delimiter-based prompts use symbols like triple quotes, brackets, or XML tags to clearly separate instruction and input.

Trainer Explanation:

Delimiters remove ambiguity, helping the AI understand exactly what content you want it to operate on.

Hands-On Exercise:

Prompt: “Summarize the following text in 3 lines: """ Kubernetes is a system for managing containerized applications across clusters… """”

Expected Outcome:

A clean and accurate summary extracted only from the delimiter content.

Trainer Explanation:

Delimiters help the model understand exactly what text you want it to operate on.

Hands-On Exercise:

Prompt: “Summarize the following text in 3 lines: """ Kubernetes is a system for managing containerized applications across clusters… """”

Expected Outcome:

A clean summary without confusion.

9️⃣Output-Formatting Prompt – Definition + Hands-On

Definition:

Output-formatting prompts tell the model to generate structured outputs such as JSON, tables, YAML, bullet lists, or diagrams.

Trainer Explanation:

This is especially useful for developers, DevOps engineers, analysts, and automation workflows.

Hands-On Exercise:

Prompt 1: **“Create a JSON object for an employee with name, id, role, and department.”**

Prompt 2: **“Create a table comparing Docker and Kubernetes with columns: Purpose, Use Case, Difficulty.”**

Prompt 3: **“Create a Kubernetes Deployment YAML for a Python app running on port 5000.”**

Expected Outcome:

A well-formatted JSON object, table, and YAML file.

Trainer Explanation:

These prompts shape the output into specific formats.

Hands-On Exercise:

Prompt 1: **“Create a JSON object for an employee with name, id, role, and department.”**

Prompt 2: **“Create a table comparing Docker and Kubernetes with columns: Purpose, Use Case, Difficulty.”**

Prompt 3: **“Create a Kubernetes Deployment YAML for a Python app running on port 5000.”**

Expected Outcome:

Well-structured JSON, table, and YAML outputs.
